Table 1: Literature Matrix

Paper	**Focus and Objective**	**Modeled Parameters**	**Smart Contract Function & Stage**	**Carbon Penalty**
Lee et al., 2025	PBFT algorithm for reliable charging information sharing and EVSP cooperation	Location; Capacity; Charging price; Wait time; State; Charging amount	Reward Distribution (Operational Stage)	-
Guo et al., 2024	DAG-blockchain double-auction mechanism to assign EVs to CSs efficiently	Number of charging piles; Location; Efficiency; Bids; Charging amounts	Trading and Assignment (Operational Stage)	-
Teimoori et al., 2022	Vertical Federated Learning algorithm for private charging station recommendation	Cost; Charging capacity; Location; Fees; Wait time; Available spots	Node Authentication (Operational Stage)	-
Hamed et al., 2025	FOA-B algorithm for optimizing CS localization and secure EV routing	Installation & maintenance cost; Load; Grid capacity; EV battery degradation	Transaction Verification (Operational Stage)	-
Sun et al., 2023	PBFT consortium blockchains for shared EV charging fee calculation and recommendation	Utilization rate; Unit price; Reputation; Wait time; Location distance	Fee Calculation and Recommendation (Operational Stage)	-
Bhawana et al., 2024	Knapsack optimization and consortium blockchain for EV renewable energy allocation	Station capacity; Smart meter readings; Renewable energy price; Surplus charges	Energy Trading and Allocation (Operational Stage)	-
Husain et al., 2025	Repeated Single-Minded Bidder auction on smart contracts for mobile charging allocation	Charging points quantity; Energy capacity; Remaining state of charge; Energy price	Allocation and Payment (Operational Stage)	-
S. M. Danish et al., 2021	Decentralized decision optimization problem for efficient EV charging station selection	Charging price; Parking cost; Charging power; Waiting time; Station reputation	Reservation and Authentication (Operational Stage)	-
Javed et al., 2022	Consortium blockchain and matching pools algorithm for EV energy trading coordination	Current state of charge; Location; Queued EVs; Battery capacity	Trade Matching and Payment (Operational Stage)	-
Al-Shehari et al., 2024	Mayfly Pelican Optimization Algorithm for secure energy and data trading	Trading energy volume; Trading rate; Transaction cost; Memory usage	Account Mapping and Trading (Operational Stage)	-

Paper	Focus and Objective	Modeled Parameters	Smart Contract Function & Stage	Carbon Penalty
Y. Wang et al., 2022	Joint coalition-matching game and reputation-based secure PCP sharing for EVs	Available solar energy; Charging price; Charging outlets; Distance; Energy demand	Energy Sharing (Operational Stage)	-
Tanwar et al., 2022	Double-auction mechanism with IPFS storage to optimize EV charging payoffs	Location; Distance; Willing prices; Energy level; Charging station prices	Reservation and Pricing (Operational Stage)	-
Xu et al., 2022	OCIO and mPSO algorithms for EV charging and renewable energy dispatching	Charging power; Storage capacity; Charging efficiency; Electricity price; State of charge	Power Dispatching (Operational Stage)	-
Augello et al., 2023	Smart contracts for certifying EV battery State of Health and V2G usage	Charging power; Ambient temperature; Cell voltage; SOC; Battery State of Health	SOH Certification (Second-Life Stage)	-
Husain et al., 2025	Context-aware machine learning and RSMB auction for mobile EV charging allocation	Battery capacity; Requested energy; Charging time; Location coordinates; Travel cost	Auction and Allocation (Operational Stage)	-
Moschella et al., 2021	DLT architecture and feedback controller for decentralized EV-CS assignment and compliance	Distance; Energy required; Charging price; Queue length; Waiting time	Compliance Enforcement (Operational Stage)	-
M. Moniruzzaman et al., 2024	RL and PoMCS blockchain for EV to Mobile Charging Station energy trading	Battery capacity; Energy price; Distance; State of charge; Trading amount	Transaction Validation (Operational Stage)	-
Rajaram and Almalki, 2026	E2-DTwin framework using GNN and PoA blockchain for urban energy optimization	Voltage; Current; Temperature; State of charge; Power generation	Automated Settlement (Operational Stage)	Yes
Husain et al., 2026	Matching game and RSMB auction for EV coalitions and mobile charging allocation	Number of charging points; Capacity; Energy demand; Location; EV reputations	Allocation and Reputation (Operational Stage)	-
Fu et al., 2021	Multi-agent simulation and blockchain for peer-to-peer private charging pile sharing	Vacant time; Electricity price; Service price; Location; Pile ID	Payment and Verification (Operational Stage)	-
Chang et al., 2025	ADMM-based decentralized P2P energy-backed token market optimization	Generation; Demand; Charging rates; Capacity; State of charge; Energy price	Market Settlement and Demurrage (Operational Stage)	-

Paper	Focus and Objective	Modeled Parameters	Smart Contract Function & Stage	Carbon Penalty
C. Chen et al., 2026	DAG-based blockchain to incentivize V2X popular content dissemination and secure charging	Energy amount; Time windows; Cryptographic proofs; Charging price	Dissemination and Payment (Operational Stage)	-
Barnawi et al., 2023	Blockchain-based demand response game theory for V2G energy trading	State of charge; Energy required; Energy coins; Power capacity; Processing power	Authentication and Reward (Operational Stage)	-
Li et al., 2023	Non-myopic shared charge post matching strategy via Ethereum smart contracts	Charge post location; Idle time; Charging price; Supported power; Distance	Match and Authentication (Operational Stage)	-
Kavitha and Sharmila, 2025	Smart BMS using XGBoost, GWO scheduling, and permissioned blockchain with HE	SOC; SOH; Voltage; Current; Temperature; Charging cost; Travel distance	Scheduling and Payment (Operational Stage)	-
Kumar et al., 2025	TAURITE framework using Stackelberg equilibrium for demand-supply balance in blockchained energynet	Distance; Speed; Charge holding capacity; Queue length; Charging capacity	Key Generation and Allocation (Operational Stage)	-
Cai et al., 2024	TEMP two-stage energy management using blockchain, ACO, and DRL for cost-effective V2V/V2G	Location; Energy price; Storage level; Parking availability; Distance; Battery degradation	Match and Dispatch (Operational Stage)	-
Choubey et al., 2024	SaPPoA consensus and IPFS-based blockchain for scalable V2V energy trading and super-stability	Velocity; State-of-charge; Required power; Battery health; Waiting time	Matching and Verification (Operational Stage)	-
Chen et al., 2024	ISAT framework with fast-path PBFT and Stackelberg game for trustworthy V2G scheduling	State of charge; Energy quantity; Charging & discharging price; Battery degradation; Load	Pricing and Settlement (Operational Stage)	-

Table 3: 中文

Paper	**焦点与目标**	**建模参数**	**智能合约功能与阶段**	**碳惩罚**
Lee et al., 2025	用于可靠共享充电信息和电动汽车服务提供商(EVSP)合作的PBFT（实用拜占庭容错）算法	位置；容量；充电价格；等待时间；状态；充电量	奖励分配（运营阶段）	忽略
Guo et al., 2024	用于将电动汽车高效分配给充电站(CS)的DAG（有向无环图）区块链双向拍卖机制	充电桩数量；位置；效率；出价；充电量	交易与分配（运营阶段）	忽略
Teimoori et al., 2022	用于私人充电站推荐的垂直联邦学习算法	成本；充电容量；位置；费用；等待时间；可用空位	节点认证（运营阶段）	忽略
Hamed et al., 2025	用于优化充电站选址和电动汽车安全路线规划的FOA-B（基于区块链的狐狸优化）算法	安装/维护成本；负荷；电网容量；电动汽车电池退化	交易验证（运营阶段）	忽略
Sun et al., 2023	用于共享电动汽车充电费用计算和推荐的PBFT联盟链	利用率；单价；信誉；等待时间；位置距离	费用计算与推荐（运营阶段）	忽略
Bhawana et al., 2024	用于电动汽车可再生能源分配的背包优化和联盟链	充电站容量；智能电表读数；可再生能源价格；剩余电量费用	能源交易与分配（运营阶段）	忽略
Husain et al., 2025	智能合约上用于移动充电分配的重复单目标竞标者(RSMB)拍卖机制	充电点数量；电量容量；剩余荷电状态(SoC)；能源价格	分配与支付（运营阶段）	忽略
S. M. Danish et al., 2021	用于高效选择电动汽车充电站的分散式决策优化问题	充电价格；停车成本；充电功率；等待时间；充电站信誉	预订与认证（运营阶段）	忽略
Javed et al., 2022	用于电动汽车能源交易协调的联盟链和匹配池算法	当前荷电状态(SoC)；位置；排队的电动汽车；电池容量	交易匹配与支付（运营阶段）	忽略
Al-Shehari et al., 2024	用于安全能源和数据交易的蜉蝣鹈鹕优化算法(MPOA)	交易能源量；交易率；交易成本；内存使用量	账户映射与交易（运营阶段）	忽略
Y. Wang et al., 2022	针对电动汽车的联合联盟匹配博弈和基于信誉的安全私人充电桩(PCP)共享	可用太阳能；充电价格；充电插座；距离；能源需求	能源共享（运营阶段）	忽略
Tanwar et al., 2022	采用IPFS（星际文件系统）存储以优化电动汽车充电收益的双向拍卖机制	位置；距离；意愿价格；电量水平；充电站价格	预订与定价（运营阶段）	忽略

Paper	焦点与目标	建模参数	智能合约功能与阶段	碳惩罚
Xu et al., 2022	用于电动汽车充电和可再生能源调度的OCIO（综合成本和满意度目标函数）和mPSO（改进的粒子群优化）算法	充电功率；储能容量；充电效率；电价；荷电状态(SoC)	电力调度（运营阶段）	忽略
Augello et al., 2023	用于认证电动汽车电池健康状态(SOH)和V2G（车对电网）使用的智能合约	充电功率；环境温度；电池电压；荷电状态(SOC)；电池健康状态(SOH)	电池健康状态认证（梯次利用阶段）	忽略
Husain et al., 2025	用于移动电动汽车充电分配的上下文感知机器学习和RSMB拍卖机制	电池容量；请求电量；充电时间；位置坐标；出行成本	拍卖与分配（运营阶段）	忽略
Moschella et al., 2021	用于分散式电动汽车-充电站分配和履约的DLT（分布式账本技术）架构与反馈控制器	距离；所需电量；充电价格；排队长度；等待时间	履约强制执行（运营阶段）	忽略
M. Moniruzzaman et al., 2024	用于电动汽车向移动充电站进行能源交易的RL（强化学习）和PoMCS（移动充电站证明）区块链	电池容量；能源价格；距离；荷电状态；交易量	交易验证（运营阶段）	忽略
Rajaram and Almalki, 2026	使用GNN（图神经网络）和PoA（权威证明）区块链进行城市能源优化的E2-DTwin（高能效去中心化数字孪生）框架	电压；电流；温度；荷电状态；发电量	自动结算（运营阶段）	Yes
Husain et al., 2026	用于电动汽车联盟和移动充电分配的匹配博弈和RSMB拍卖机制	充电点数量；容量；能源需求；位置；电动汽车信誉	分配与信誉（运营阶段）	忽略
Fu et al., 2021	用于点对点(P2P)私人充电桩共享的多智能体模拟和区块链	空闲时间；电价；服务价格；位置；充电桩ID	支付与验证（运营阶段）	忽略
Chang et al., 2025	基于ADMM（交替方向乘子法）的去中心化点对点(P2P)能源代币市场优化	发电量；需求量；充电率；容量；荷电状态；能源价格	市场结算与滞期费管理（运营阶段）	忽略
C. Chen et al., 2026	用于激励V2X（车联网）热门内容分发和安全充电的DAG区块链	电量；时间窗；密码学证明；充电价格	内容分发与支付（运营阶段）	忽略
Barnawi et al., 2023	基于区块链的需求响应博弈论，用于V2G能源交易	荷电状态；所需电量；能源币；电力容量；处理算力	认证与奖励（运营阶段）	忽略
Li et al., 2023	基于以太坊智能合约的非近视共享充电桩匹配策略	充电桩位置；空闲时间；充电价格；支持功率；距离	匹配与认证（运营阶段）	忽略
Kavitha and Sharmila, 2025	采用XGBoost、GWO（灰狼优化算法）调度以及带HE（同态加密）许可链的智能BMS（电池管理系统）	荷电状态(SOC)；健康状态(SOH)；电压；电流；温度；充电成本；行驶距离	调度与支付（运营阶段）	忽略

Paper	焦点与目标	建模参数	智能合约功能与阶段	碳惩罚
Kumar et al., 2025	在区块链能源网络中使用斯塔克尔伯格(Stackelberg)均衡实现供需平衡的TAURITE框架	距离；速度；电池保持容量；排队长度；充电容量	密钥生成与分配（运营阶段）	忽略
Cai et al., 2024	利用区块链、ACO（蚁群优化）和DRL（深度强化学习）实现高性价比V2V/V2G的TEMP两阶段能源管理	位置；能源价格；储能水平；停车可用性；距离；电池退化	匹配与调度（运营阶段）	忽略
Choubey et al., 2024	用于可扩展V2V能源交易及实现超稳定性的基于SaPPoA共识与IPFS的区块链	速度；荷电状态(SOC)；所需功率；电池健康；等待时间	匹配与验证（运营阶段）	忽略
Chen et al., 2024	采用快速路径PBFT和斯塔克尔伯格(Stackelberg)博弈实现可信V2G调度的ISAT框架	荷电状态(SOC)；能量；充放电价格；电池退化；负荷	定价与结算（运营阶段）	忽略